

activities such as walking, sitting, cooking, or exercising allows devices to adjust lighting, temperature, and entertainment settings dynamically. For example, if a person begins a workout, the system can automatically brighten the room and play an energetic playlist. Similarly, if an individual is detected sleeping, the system can dim the lights and turn off unnecessary appliances to save energy.

Despite its advantages, WiFi-based motion recognition has faced a key challenge – environmental interference. WiFi signals fluctuate due to changes in furniture placement, human presence, and electronic interference, leading to inconsistent recognition accuracy. The new MSF-Net framework tackles this issue with a sophisticated deep-learning approach that refines motion detection to an unprecedented level.

How MSF-Net improves accuracy and efficiency

Professor Jeon and his team designed MSF-Net to enhance WiFi-based activity recognition by integrating multiple advanced AI techniques. The framework consists of three primary components:

1. **A Dual-Stream Structure** – This component utilizes short-time Fourier transform (STFT) and discrete wavelet transform (DWT) to analyze channel state information (CSI) from WiFi signals. This dual-stream approach helps pinpoint irregularities and fluctuations in signal patterns caused by human movement.
2. **A Transformer Model** – The transformer extracts high-level features from CSI data, ensuring that motion detection remains robust even in dynamic environments where furniture layouts or human movement patterns change frequently.
3. **An Attention-Based Fusion Branch** – This fusion mechanism combines data from multiple streams to improve recognition accuracy further. By integrating different perspectives of motion data, the system reduces errors and enhances real-time responsiveness. The effectiveness of this approach



was validated through extensive testing on well-established datasets, including SignFi, Widar3.0, UT-HAR, and NTU-HAR. MSF-Net achieved Cohen's Kappa scores of 91.82%, 69.76%, 85.91%, and 75.66% on these datasets, respectively – outperforming existing methods in WiFi-based activity recognition.

What's beyond smart homes?

For consumers, this breakthrough means that AIoT-powered homes will become significantly smarter and more adaptive. Imagine walking into a room and having the environment adjust itself instantly based on your activity without needing voice commands or manual inputs. Whether it's adjusting the room temperature when you go to bed or detecting a fall and alerting emergency services, the applications of MSF-Net extend far beyond convenience.

One of the most promising areas of application is healthcare and elder care. With an aging global population, AIoT-enabled homes equipped with MSF-Net can offer unobtrusive monitoring for seniors, ensuring their safety while respecting their privacy. Instead of relying on cameras or wearable devices, the system can detect irregular movements, such as a fall, and trigger alerts for caregivers or emergency responders. This can significantly reduce response times in critical situations and improve overall well-being.

In addition to healthcare, the new technology can also revolutionize home

security. Traditional security systems rely on motion sensors that may trigger false alarms due to pets or environmental changes. MSF-Net, however, offers a much more precise recognition of human movement, reducing the likelihood of false positives while ensuring enhanced security.

Furthermore, energy efficiency in smart homes can see a major boost with this advancement. AIoT systems powered by MSF-Net can optimize power consumption by detecting when a room is unoccupied and automatically turning off lights and appliances. In a time when sustainability and energy conservation are major priorities, this could lead to significant cost savings and a reduced carbon footprint for households worldwide.

A smarter, more adaptive future

Despite the impressive performance of MSF-Net, challenges remain in the widespread adoption of WiFi-based motion recognition. Environmental variations such as interference from multiple WiFi sources, furniture placement, and different home layouts can still impact accuracy. However, continuous improvements in AI and deep learning models will likely refine these aspects over time.

Additionally, there are concerns about the computational power required for real-time processing. While AI-driven WiFi recognition is an exciting concept, it demands substantial processing capabilities. Future iterations of MSF-Net may integrate edge computing to distribute processing tasks efficiently, ensuring faster and more efficient recognition without overloading home networks.

Moreover, consumer adoption will depend on seamless integration with existing smart home ecosystems. Leading smart home companies such as Google, Amazon, and Apple may need to develop standardized frameworks that allow MSF-Net-powered recognition systems to work effortlessly with their smart assistants and devices. **d**